

Experimental and two-scale numerical studies on the behavior of prestressed concrete-steel hybrid wind turbine tower models

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ABSTRACT

Prestressed concrete-steel hybrid (PCSH) wind turbine tower has been recognized as an efficient alternative to traditional steel tube supporting towers for wind farms in harsh sites such as mountain regions. To investigate the global and local mechanical behavior including failure patterns of a full scale PCSH wind turbine towers, an experimental study on a substructure of a scaled PCSH tower is carried out firstly. Then, a material subroutine for three-dimensional (3D) fiber beam elements suitable for implicit analysis on the mechanical behavior of the substructure of the scaled PCSH tower is developed. A two-scale numerical model of the substructure of the scaled PCSH wind turbine tower employing the developed 3D fiber beam elements and solid elements is developed. The accuracy and efficiency of the developed two-scale numerical model for the behavior simulation of the substructure of the scaled PCSH wind turbine tower is illustrated by comparing the numerical results with the experimental measurements. Additional comparison of the simulation results using the developed two-scale numerical model is made with them of a model using sole 3D fiber beam elements and a model using sole solid elements, respectively. Results show that the force–displacement curve determined with the developed two-scale model is close to the test result and coincident well with that of the two models using 3D fiber beam elements and the solid elements. The developed two-scale model can describe the global behavior, the local failure patterns and the stress distribution of the tested specimen accurately with improved simulation efficiency. Finally, the validated two-scale modelling approach is employed to analyze the static behavior, modal parameter, and fatigue of the optimized full-scale PCSH wind turbine tower previously developed by the authors. The simulation results show that the two-scale model can reflect the stress of the PCSH wind turbine tower at the critical position as well as the global mechanical behavior with higher computational efficiency. The fiber beam model can be used for predicting only global, but not local, behavior.

1. Introduction

Given the trend towards developing wind turbines with higher hub elevation in harsh construction sites such as mountain regions, it is necessary to develop novel and efficient supporting systems for wind turbines with better static and dynamic properties, transportation and construction convenience and economic advantages [1]. The application of traditional steel tube towers with large diameters for large capacity wind turbines in mountain regions is limited due to the difficulty and risk in transportation and extra investment in temporary road construction [2]. As an alternative of traditional steel tubular towers,

prestressed concrete (PC) towers can be either cast on site or precast in a factory and then assembled on site in pieces. The PC towers usually have better dynamic behavior and global and local stability when compared with traditional steel tubular towers. Ercan et al. [3] presented a concept of modular prefabricated post-tensioned concrete wind turbine towers which is testified by numerical simulation. Cao et al. [4] experimentally and numerically studied the seismic performance of prestressed segmental concrete wind turbine towers. In recent years, as a new alternative of supporting structure for wind turbine, prestressed concrete-steel hybrid (PCSH) wind turbine tower has shown its advantages including better static and dynamic behavior, convenience in

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transportation, longer service life and less costs in the form of the levelized cost of energy (LCOE) [5,6].

Numerical simulation on the behavior of the supporting system of a modern wind turbine is usually a time-consuming and cost-inefficient task because of the massive number of finite elements required [7]. Moreover, to properly understand the behavior of different type of wind turbine towers, it is of significance to take both geometric and material nonlinearity into account. At present, most of the dynamic behavior analysis for PC wind turbine towers is carried out by simplifying the wind turbine towers into multi-degree-of-freedom (MDOF) models or converting the dynamic problem into a static problem. Harte and Basu [8] developed an MDOF model for a horizontal axis wind turbine supporting structure to investigate its dynamic behavior using an Euler-Lagrangian approach. Ma et al. [9] provided an optimization design approach for a PC wind turbine tower where the dynamic loads such as wind and earthquake are converted into static loads. However, the shortages of the employed MDOF models are the simulated global behavior is not accurate enough and the local failure pattern of the wind turbine towers cannot be described with acceptable accuracy. Seismic response analysis of a PC wind turbine tower is carried out adopting a static load determined by the response spectrum [10]. To properly investigate the behavior of PC towers including material and geometric non-linearity, Kenna and Basu [11] suggested a finite element method (FEM) employing shell elements and spring elements with large stiffness, which are used to constrain the translation degree-of-freedom (DOF) between the truss elements for prestressed reinforcements and shell elements for concrete. To model the wind-induced response of wind turbine tower, Wang and Wu [12] developed a knowledge-enhanced deep learning (KEDL) algorithm with high accuracy and efficiency. Zhao et al. [13] introduced a fast convolution integration method to accelerate the computation of the dynamic analysis of non-proportionally damped structures. In practice, traditional FEM analyses for full-scale wind turbine towers with solid elements and nonlinear constitutive laws for concrete is a challenging task because of its high computational costs. It is of importance to develop an efficient and accurate approach for the prediction of both global behavior and local failure patterns of wind turbine towers.

Multi-scale simulation provides a more reliable way to investigate structure behavior at different scales with sufficient precision and lower computational costs [14]. Chen et al. [15] revealed the detailed failure pattern and the crack evolution of a reinforced concrete (RC) bridge column interface based on two-dimensional multiscale models. Gu et al. [16] implemented the multi-scale simulation method for concrete structures by combining FEM and discrete element method. Sun et al. [17] introduced a multi-scale beam lattice method to explore mesoscopic fracture mechanisms with higher computational efficiency. Chen et al. [18] proposed a XFEM-based multiscale modeling approach to investigate the monotonic and hysteretic performance of concrete columns and reveal its cracking behavior. Moyeda and Fish [19] carried out a multiscale beam element method to predict the initial effects of prestressing, creep, shrinkage, and temperature of PC structures and verified the method by experimental measurement. Sadowski [20] explored a hybrid beam-shell modeling technique for steel tube wind turbine towers to investigate the criticality in all tower strakes. Calibrated by the measurements of an actual tower, Szafran et al. [21] established a beam finite element model with shell elements for the joints to evaluate the response of a steel lattice tower under random wind loads.

This paper aims at the experimental and two-scale numerical investigation on the mechanical behavior of PCSH wind turbine towers. In order to develop three-dimensional (3D) fiber beam elements for the two-scale models, experimental study on a substructure of a reduced-scale PCSH tower specimen is carried out at first. The test results are used to validate the 3D fiber beam elements. Combining the 3D fiber beam elements and solid elements, a two-scale finite element modelling approach is utilized to investigate both the global mechanical behavior of the PCSH wind turbine tower specimen and the local failure pattern of

critical parts with limited computation cost. In the two-scale model, the interested PC segment is simulated with solid elements and the rest of the PC segment as 3D fiber beam elements. Two material subroutines for 3D fiber beam elements for the PC segment are developed. The interaction between the concrete and the prestressing tendons is simulated by a customized connection type. The accuracy and efficiency of the developed two-scale model for the behavior investigation of the scaled PCSH tower specimen is presented by comparing the numerical results with the experimental measurements, the simulation results of the models adopting 3D fiber beam elements and solid elements only, respectively. In addition, the validated two-scale modeling approach and the developed material subroutine are adopted to investigate the mechanical behavior of an optimized full-scale PCSH wind turbine tower previously developed by the authors.

2. Experimental study on a substructure of a scaled PCSH tower

Experimental study on the mechanic behavior of a scaled PCSH wind turbine tower specimen under static loadings is carried out firstly. The experimental results of the tested scaled specimen are used to verify the two-scale model, the model with sole 3D fiber beam elements and sole solid elements.

2.1. Substructure of the scaled PCSH wind turbine model and test

In this section, a substructure of a specimen of a 77.5 m full scale PCSH wind turbine tower with geometric similarity ratio of 0.1 is investigated, as shown in Fig. 1 [22]. The dimensions of the prototype tower and the parameters of the wind turbine are respectively listed in Table 1 and Table 2. The area of the reinforcement and prestressed tendons are 47,100 mm² and 48,020 mm², respectively. Two sets of load combinations are considered in the prototype tower design. One corresponding to the serviceability limit state (SLS) has a load factor of 1.0 for all load categories. The other corresponds to the ultimate limit state (ULS) and its load factors are equal to 1.0 and 1.4 for permanent and wind loads, respectively. The scaled PCSH tower specimen is shown in Fig. 1(a)-(c). The substructure of the specimen has a strong foundation and is anchored on the strong floor of the lab. The top of the specimen is connected to an electro-hydraulic servo control actuator to apply horizontal loading. The dimensions of the steel and PC segment are shown in Fig. 1(d) and (e), respectively. The PC segment and the steel segment are connected by a circular steel flange with a thickness of 25 mm. The load acting on the specimen can be calculated based on the parameters of the prototype tower and the corresponding references [23]. The vertical prestressing force of 544.8 kN is applied on the concrete tube with six Φ^S 1 $\times$ 7 prestressed tendons of 12.7 mm diameter. The axial force of 42.3 kN was applied to the top of the specimen, which is determined by the similarity ratio of the specimen to the original full scale PCSH tower. To simulate the actual stress state of the PC segment of the substructure of the PCSH tower under the SLS and ULS, the loading procedure is divided into three stages divided by two target lateral forces of 24.1 kN and 34.5 kN, respectively [23]. Horizontal displacement control is used during the loading process, and the loading rate is 0.5 mm/min. As presented in Fig. 1 (f), eight dial gauges are arranged at the heights of 140 mm, 470 mm, 780 mm, 1,050 mm, 1,550 mm, 1,620 mm, 2,120 mm and 2,400 mm above the base, respectively. The horizontal reaction force can be measured from the actuator directly. The test loading device is illustrated in Fig. (g). A 50 t force sensor is arranged on the top of the specimen. The measured average compressive strength for the concrete is 48.9 MPa. The ultimate strength of 6 and 8 reinforcements are 530.98 MPa and 526.85 MPa, respectively.

2.2. Experimental observation

No collapse of the steel segment and no separation between the flange and the PC segment is observed during the whole loading

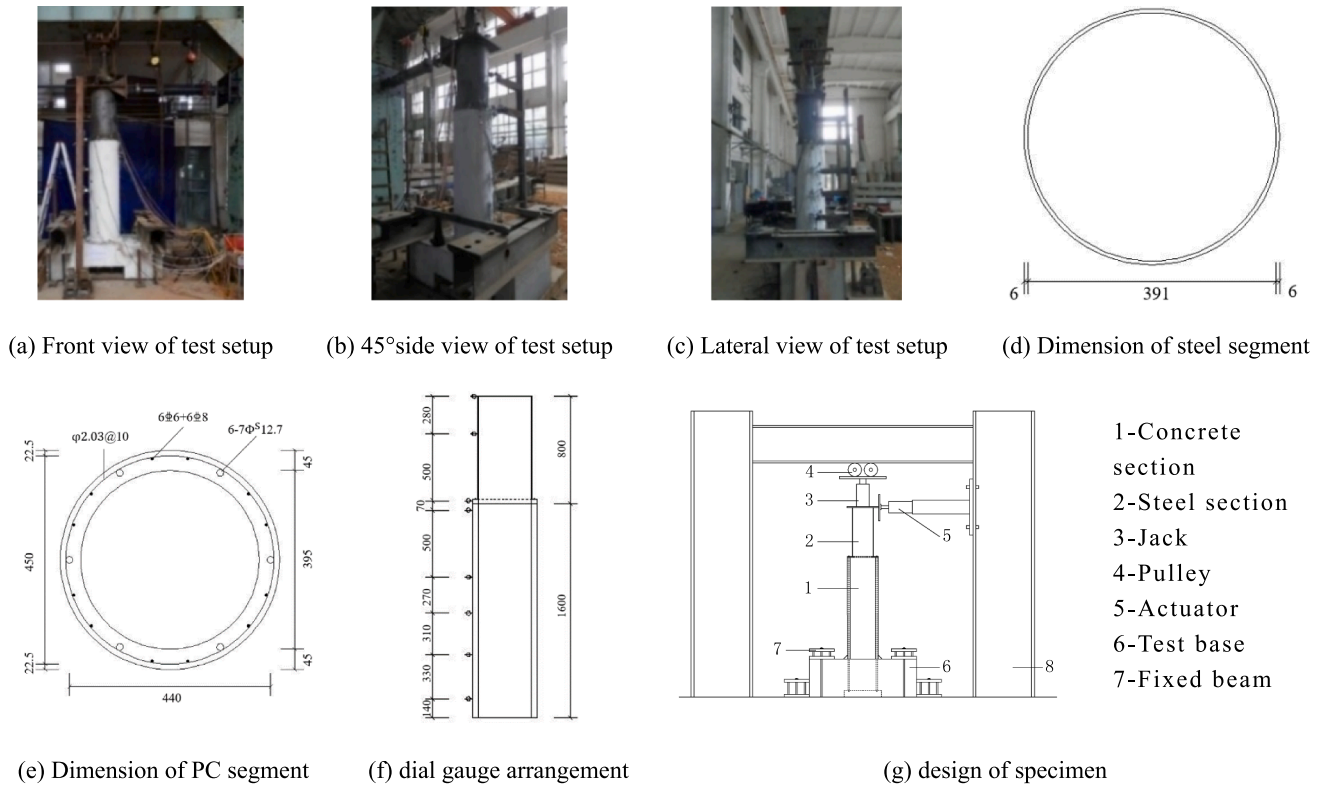


Fig. 1. Test setup.

Table 1
Parameters of the full scale PCSH wind turbine tower.

Segment	Height (m)	Thickness (mm)	Top diameter (mm)	Bottom diameter (mm)
Steel segment 3	21.5	14	2,686	3,485
Steel segment 2	20	18	3,485	4,046
Steel segment 1	20	20	4,046	4,400
PC segment	16	450	4,400	4,400

Table 2
Parameters of the wind turbine.

Item	Rated Power (MW)	Number of blades	Rotor diameter (m)	Wind turbine mass (t)	Rated wind speed (m/s)	Cut-out wind speed (m/s)
Value	2	3	93.4	128.5	11	25

procedure. There is no crack in the base foundation and the concrete part at the connection between concrete and steel flange during the test. The specimen top displacement versus top horizontal load is shown in Fig. 2. According to Fig. 2, the structural stiffness does not have obvious change until the top displacement reached 6.84 mm, then decreases slightly. The load-carrying capacity of the model in the test is greater than the equivalent value under the ULS and no cracks are detected in the factored SLS. It can be observed that the structure can provide sufficient strength in the ULS and the SLS. Cracking occurred at the lower part of the PC segment when the top displacement reached 9.58 mm and the horizontal force reached 48.95kN which exceed the load under the ULS. As shown in Fig. 3, crack develops in the horizontal direction, indicating that the main damage mode of the structure is bending

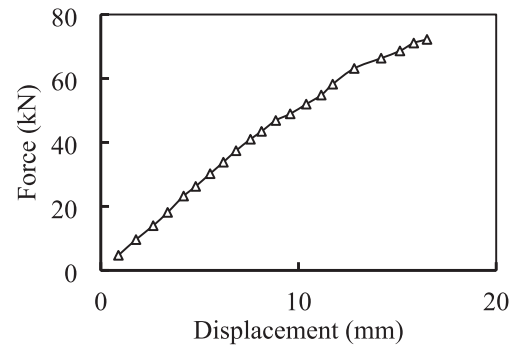


Fig. 2. Load-displacement curve.



Fig. 3. Flexural cracks in the specimen.

damage.

3. Two-scale mode and simulation for the substructure of the scaled PCSH tower

To investigate the global and local behavior including failure patterns of the substructure of the scaled PCSH wind turbine tower specimen, a two-scale finite element modeling approach is proposed by modeling the lower selected concrete parts with higher stress level with fine solid elements and the rest of the substructure of the scaled specimen with beam elements. Maximum stress and damage of the concrete segment of the PCSH wind turbine tower often occurs at the bottom of the tower or the connection between the concrete and steel flange of the PCSH substructure where the maximum stress is likely to be located. Therefore, solid elements are employed to model the upper and lower parts of the concrete segment, and 3D fiber beam elements are applied to simulate the rest part of the wind turbine support in the two-scale model.

3.1. Material model for the substructure of the PCSH wind turbine tower

3.1.1. 3D fiber beam

Fiber beam element has been universally applied to simulate the nonlinear behavior of composite cross-sections. As shown in Fig. 4, the reinforcement is equivalent to a tubular section with the same area and moment of inertia to assign the properties of the reinforcement section to the beam element.

In Fig. 4, d is the diameter of the concrete section, c is the thickness of the protective layer, t is the thickness of the equivalent circular section of the reinforcement, A_s and I are the area and the moment of inertia of the reinforcement, respectively.

In the 3D fiber beam element, the bilinear constitutive model with a reverse loading curve pointing to the historical maximum deformation point proposed by Clough is used to model the reinforcement and the hysteresis constitutive model of Concrete02 in Opensees is used as constitutive law of concrete. The constitutive laws for both reinforcement and concrete are shown in Fig. 5. The user-defined material (UMAT) for 3D fiber beam elements has been developed and used as the material subroutines of ABAQUS which is used as a solver in this study [24–26].

In Fig. 5(a), α and λ take the value of 0.04 and 0.3, respectively. E_0 is the initial stiffness of reinforcement and f_y is the yielding strength of steel reinforcement.

The constitutive law of concrete shown in Fig. 5(b) is described by the following equations:

$$\sigma_c = \begin{cases} f_{c0} \frac{\epsilon_c}{\epsilon_{c0}} \left(2 - \frac{\epsilon_c}{\epsilon_{c0}} \right) & 0 < \epsilon_c < \epsilon_{c0} \\ f_{c0} + \frac{f_{cu} - f_{c0}}{\epsilon_{cu} - \epsilon_{c0}} (\epsilon_c - \epsilon_{c0}) & \epsilon_{c0} < \epsilon_c < \epsilon_{cu} \\ f_{cu} \epsilon_c > \epsilon_{cu} \end{cases} \quad (1)$$

$$\epsilon_r = \frac{f_{cu} - E_{cu} \epsilon_{cu}}{E_c - E_{cu}} \quad (2)$$

$$\sigma_r = E_c \epsilon_r \quad (3)$$

$$E_c = \frac{2f_{c0}}{\epsilon_{c0}} \quad (4)$$

$$\sigma_t = \begin{cases} E_c \epsilon \leq \epsilon_t \\ f_t - E_t \left(\epsilon - \frac{f_t}{E_c} \right) & \epsilon > \epsilon_t \end{cases} \quad (5)$$

where f_{c0} is the axial compressive strength of concrete, ϵ_{c0} is the concrete strain when the stress reaches the axial compressive strength, ϵ_{cu} is the ultimate compressive strain, f_{cu} is the concrete strength at ultimate compressive strain, σ_c is the compressive stress of concrete, ϵ_c is the compressive strain of concrete, E_c is the initial stiffness, E_{cu} is the stiffness at ultimate compressive strain, f_t is the tensile strength, σ_t is the tensile stress and ϵ_t is the peak tensile strain of the concrete.

3.1.2. Concrete

The concrete damage plasticity (CDP) constitutive law shown in Fig. 6 is utilized to simulate the solid concrete elements. The curve under compression can be calculated by the Eqs (6)–(9) [27].

$$f_c = \frac{f_{cc} r \left(\frac{\epsilon_c}{\epsilon_{cc}} \right)}{r - 1 + \frac{\epsilon_c}{\epsilon_{cc}}} \quad (6)$$

$$r = \frac{E_c}{E_c - \frac{f_{cc}}{\epsilon_{cc}}} \quad (7)$$

$$f_{cc} = f_{c0} \left(-1.254 + 2.254 \sqrt{1 + \frac{7.94 f_l}{f_{c0}}} - 2 \frac{f_l}{f_{c0}} \right) \quad (8)$$

$$f_l = K_e \frac{2 f_{yh} A_{sp}}{DS} \quad (9)$$

where f_c is the compressive concrete stress, f_{cc} denotes the compressive strength of confined concrete, ϵ_c is the compressive strain of confined concrete, ϵ_{cc} is the strain at f_{cc} , the E_c is the Young's modulus of concrete, f_{c0} presents the compressive concrete stress of unconfined concrete, f_l is effective lateral confining pressure, K_e is the effective confinement coefficient of cross section, f_{yh} is yield stress of stirrups, A_{sp}

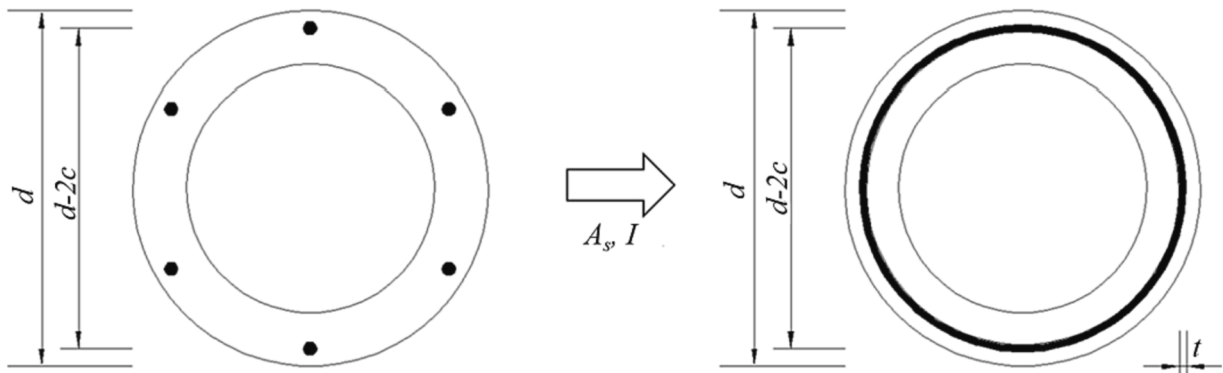


Fig. 4. Longitudinal reinforcement and section equivalence.

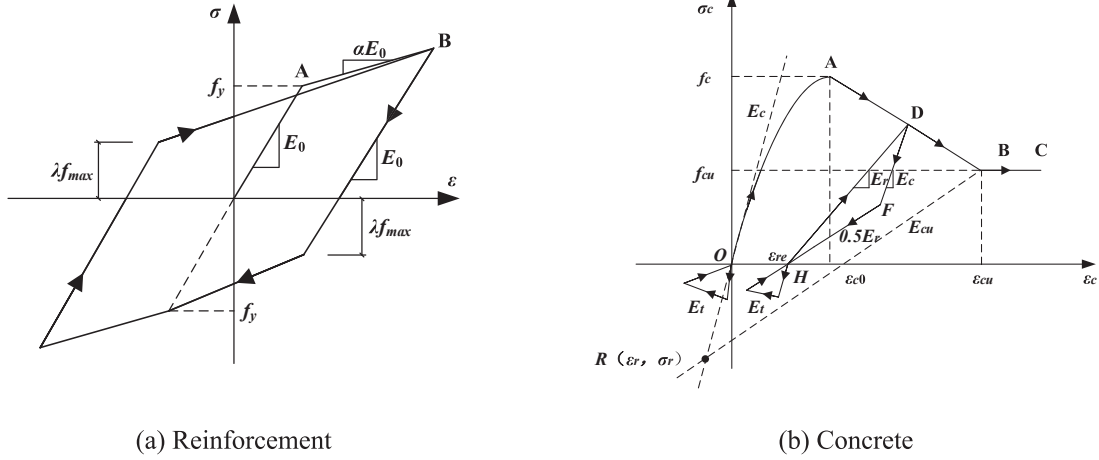


Fig. 5. Constitutive models of concrete and reinforcement for the 3D fiber beam element.

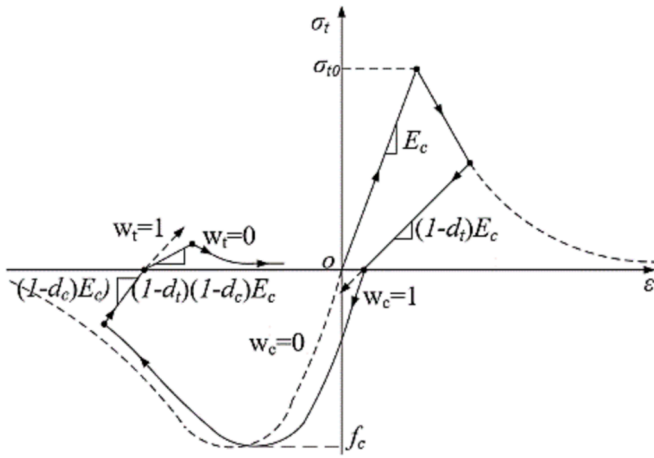


Fig. 6. The loading and unloading rules for uniaxial load cycle.

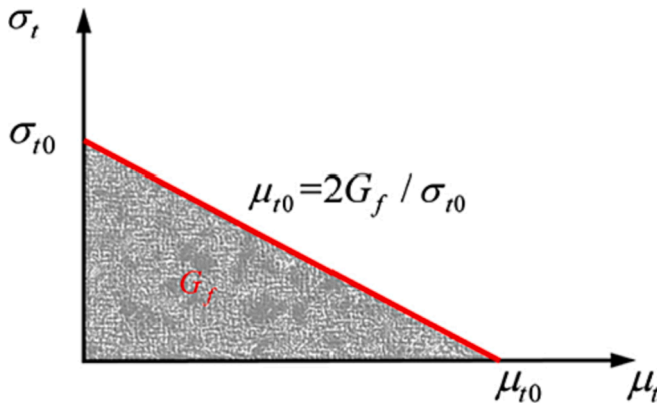


Fig. 7. Post-failure stress-fracture energy curve.

means the area of the stirrup section, D and S are the diameter and spacing of the stirrups, respectively, w_c and w_t are the compressive and tensile stiffness recovery factor, respectively.

Instead of the tensile stress-strain curve, the fracture energy presented in Eqs (10) and Fig. 7 is used to alleviate the mesh sensitivity problem[28].

$$\mu_{t0} = 2G_f / \sigma_{t0} \quad (10)$$

in which μ_{t0} stands for the crack normal displacement when the tensile stress σ_t is completely lost, σ_{t0} is the failure stress, G_f represents the fracture energy.

3.1.3. Other

The elastic-perfectly plastic stress-strain relationship is utilized to model the steel segment and the bilinear model for prestressed tendon and steel reinforcement in the segment with solid elements.

3.2. Elements

In this study, three different models including the solid element model, two-scale element model and fiber beam model are considered. The employed elements in each FEM model are listed in Table 3[29]. The B31 element is adopted to simulate the PC segment modeled by the 3D fiber beam element in the two-scale element model and the C3D8R and T3D2 element are used to simulate the concrete and reinforcement of the rest part of the PC segment, respectively.

3.3. Boundary and interaction setting

For the two-scale element model, the 3D fiber beam elements and the solid elements are connected by coupling based on the plane-section assumption. Since the prestressing force applied on the flange can prevent the separation of steel and concrete, the steel flange and concrete are directly tied together for simplicity. According to the bond-slip effect described by Kwak and Filippou, the nodes of steel reinforcement can be mapped directly onto the nodes of concrete occupied the same location to incorporate them into the overall stiffness of the model[30]. The

Table 3
Elements employed in different FEM models.

Models	Component	Element in ABAQUS
Solid element model	Steel	C3D8R
	Concrete	C3D8R
	Reinforcement	T3D2
	Prestressed tendon	T3D2
Two-scale element model	Flange	C3D8R
	Steel	B31
	Concrete	C3D8R/ B31
	Reinforcement	T3D2/ B31
Fiber beam model	Prestressed tendon	T3D2
	Flange	C3D8R
	Steel	B31
	Concrete	B31
	Reinforcement	B31
	Prestressed tendon	T3D2

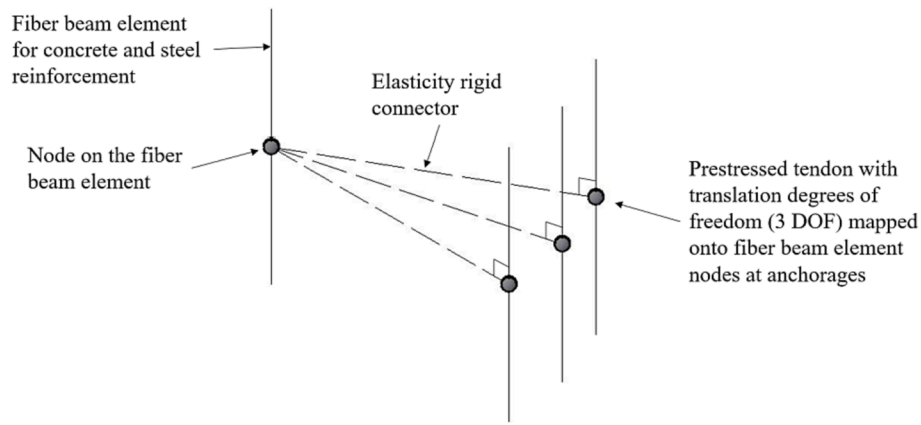


Fig. 8. Fiber beam elements with prestressed tendons overlay.

nodes on the prestressed tendons are mapped onto the nodes on the fiber beam element used as anchorage locations. As shown in Fig. 8, the elasticity rigid connector perpendicular to the tendons is utilized to restrain the translational DOF of the nodes on unbonded and prestressed tendons laterally against the nodes on the concrete segment of the tower in the two-scale element model, which do not allow the vertical load transfer between the tendons and the concrete segment.

4. Two-scale simulation for the substructure of the scaled PCSH wind turbine tower

4.1. FEM simulation on the tested specimen with different models

In this section, as illustrated in Fig. 9, the tested substructure of the scaled PCSH wind turbine tower is modeled with three different models, a solid element model, a two-scale model, and a fiber beam model for comparison. All the DOF at the bottom of the model are fixed. The horizontal force in the x-negative direction is applied on the top of the model in the form of controlled horizontal displacement. The detailed material parameters are listed in Tables 4 and 5.

In the solid element model, both the PC and the steel tube of the specimen are modeled as solid elements as shown in Fig. 9(a). Since the stress distribution of the steel tube of the PCSH specimen is relatively simply and clear and no steel tube failure is observed in the experiment procedure, the steel segment is modeled with beam elements. In the two-scale model, the top and bottom of the PC segment are modeled with solid elements and the rest of the PC segment is modeled with 3D fiber beam elements as shown in Fig. 9(b) because of the complicated stress distribution on the top and bottom parts of the specimen. In the fiber element model, both the PC and steel tube parts of the specimen are

Table 4

Material property of the steel, reinforcement, and prestressed tendon of the scaled PCSH tower.

Material	Elastic modulus (MPa)	Density (kg/m ³)	Poisson's ratio	Yield strength (MPa)	Tensile strength (MPa)
Q345	206,000	7,850	0.28	345	–
HPB300	200,000	7,850	0.3	300	420
6 rebar	200,000	7,850	0.3	393	531
8 rebar	200,000	7,850	0.3	390	527
Strand	195,000	7,850	0.3	1,320	1,860

modeled with beam elements as shown in Fig. 9(c).

4.2. Numerical simulation results of different models and their comparison

4.2.1. Comparison of the load–displacement curves of different models

The load–displacement curves of the above three models compared with the experiment result are shown in Fig. 10 and Table 6. From Fig. 10, the load–displacement curves determined by the three numerical models are close to each other and also agree with the test results. For safety, the maximum deformation of the tested substructure of the PCSH tower was 16.50 mm, which is over the displacement of the specimens under ULS. Compared with the test result, when the horizontal deformation is 16.50 mm, the relative error of the reaction force generated by the solid element model, the two-scale model and fiber beam model are 4.75 %, 6.45 % and 6.87 %, respectively. The relative errors of the three model are smaller than 10 %, which testify the accuracy of the models. As the applied displacement increases, the tangent

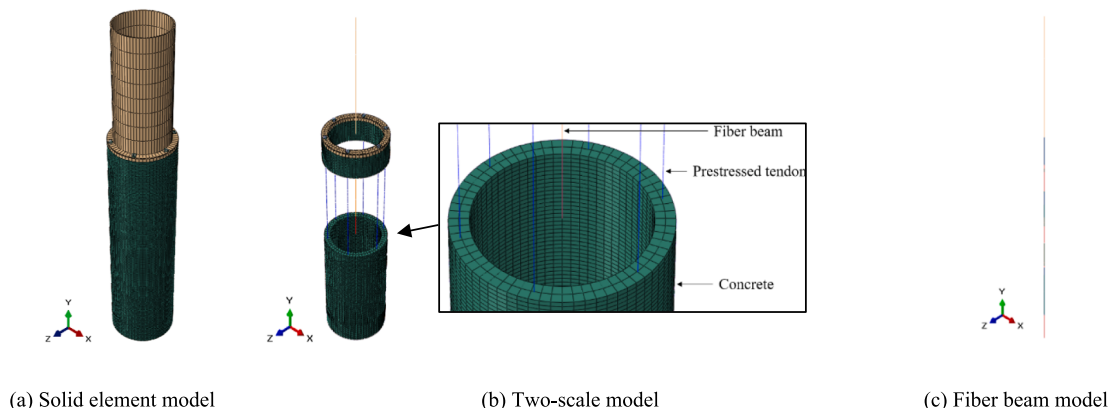
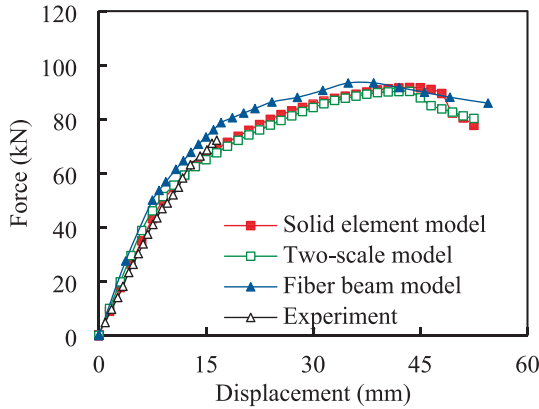


Fig. 9. Multiscale modeling approaches.

Table 5

Material property of the concrete of the scaled PCSH tower.

Material	Elastic modulus (MPa)	Density (kg/m ³)	Poisson's ratio	Compressive strength (MPa)	Tensile strength (MPa)	Dilation angles (°)	Eccentricity	K	Viscosity coefficient
C50	32,000	2,500	0.2	48.95	2.64	31	0.1	0.667	1.0e-5

**Fig. 10.** Comparison of load–displacement curves.**Table 6**

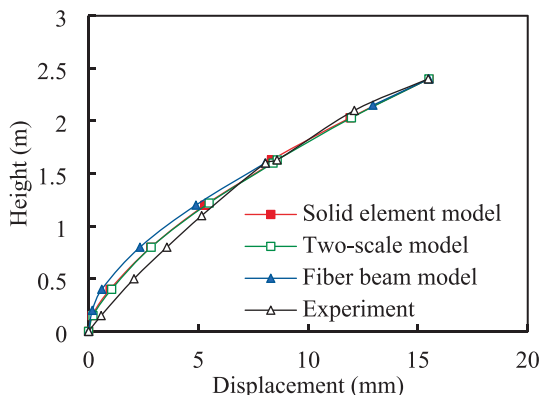
Load comparison when the horizontal deformation is 16.50 mm.

Model	Force (kN)	Relative Error (%)	Ultimate load bearing capacity (kN)	Relative Error (%)
Experiment	72.22	–	–	–
Solid element model	68.79	4.75	91.73	–
Two-scale model	67.56	6.45	90.29	1.57
Fiber beam model	77.18	6.87	93.43	1.85

stiffness of the three models decreases. The load–displacement curves of the three numerical simulation models are close, which testifies that the developed two-scale modeling approach can be employed to simulate the stiffness and ultimate load bearing capacity of the substructure of the PCSH tower structure with acceptable accuracy.

4.2.2. Deformation along the height of the tested specimen

The comparison of the deflection of the tested specimen and the three models are exhibited in Fig. 11. According to Fig. 11, the lateral displacement measurement along the height of the specimen and them

**Fig. 11.** Deflection curve.

of the three numerical models are close. The deflection curves of the three models are similar, which indicates the effectiveness of the modeling approach. The deformation error of the fiber beam model is larger when compared with them of the sole solid element model and the two-scale model.

According to Figs. 10 and 11, the two-scale model in this paper can be used to mimic the global behavior of the tested specimen with acceptable accuracy when compared with the traditional model with solid elements, the model with fiber beam elements and the experiment result. Compared with the fiber beam model, the established two-scale model is used to simulate the local behavior including the stress and the damage distribution of the tested specimen and the results are described as follows.

4.2.3. Comparison of stress distribution of PC segment of the tested specimen with different models

The tensile and compressive stress in concrete, and the stress of non-prestressed reinforcement and the prestressing tendon of the solid element model and the two-scale model when the horizontal deformation is 16.50 mm are presented in Figs. 12 and 13, respectively. The comparison of the maximum tensile and compression stress in concrete and the maximum stress in the non-prestressed reinforcement and the prestressing tendon of the above three numerical models are summarized in Table 7. The stress value and distribution of the fiber beam model can be calculated by the section forces and moments with material mechanics.

According to Figs. 12 and 13, it can be observed that the maximum stress of the reinforcement is located at the bottom of the longitudinal reinforcement at the compression side of the PC tube of the tested specimen. The maximum tensile and compressive stresses of concrete are located at the lower part of the concrete at the tensile and compression side, respectively.

According to Table 7, the relative errors between the stress of the two-scale element model and that of the model with solid elements are smaller than 5 %, which exhibits that the two-scale models can simulate the maximum structural stress at the critical location of the tested specimen with acceptable accuracy. The stress distribution and maximum stress of the model with solid elements, the two-scale models are close.

The location of the maximum stress of the fiber beam model, including maximum tensile stress in concrete, the maximum compressive stress in concrete, maximum stress of non-prestressed reinforcement, and maximum stress in prestressing tendon, is similar to that of the model with solid elements. However, the maximum stresses derived from the fiber beam model are not accurate enough and the relative error of some maximum stress of the model is greater than 10 %, especially the maximum tensile stress of concrete. This is because it is difficult to simulate the cracks using the 3D fiber beam elements. Meanwhile, the complexity and non-uniformity of the stress at the critical parts cannot be presented by the fiber beam model. Therefore, the fiber beam model cannot obtain maximum stress and local failure of the critical PC tube of the PCSH wind turbine tower accurately.

4.2.4. Comparison of damage distribution of the tested specimen with different models

The spatial distribution of the tensile and compressive damage of concrete in different models are respectively illustrated in Figs. 14 and 15 when the horizontal deformation is 16.50 mm. The compressive damage is low, and the tensile damage is mainly concentrated at the

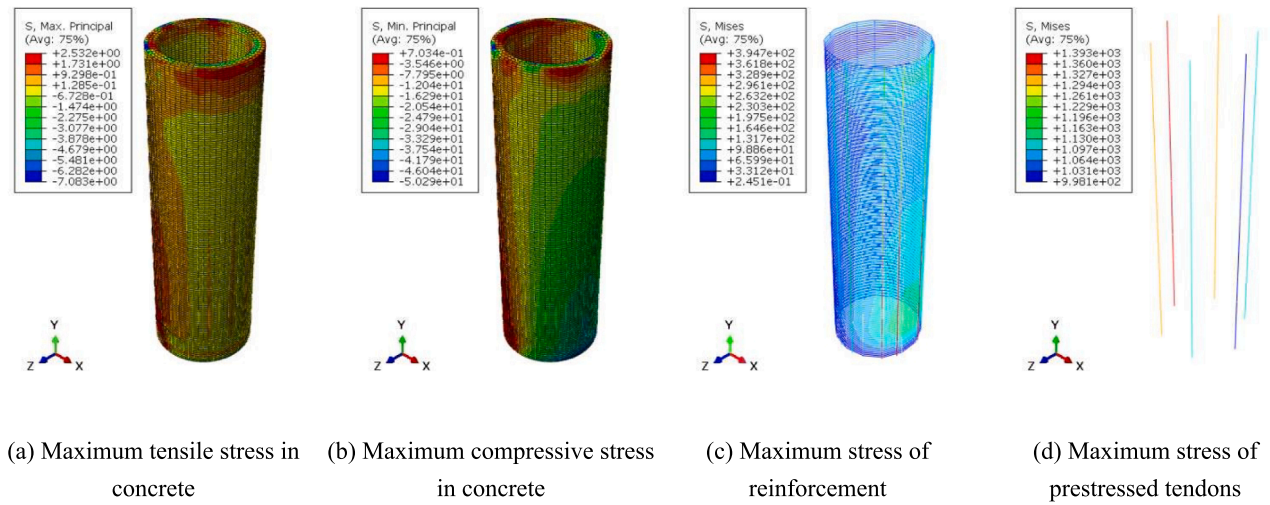


Fig. 12. Stress nephogram of the solid element model.

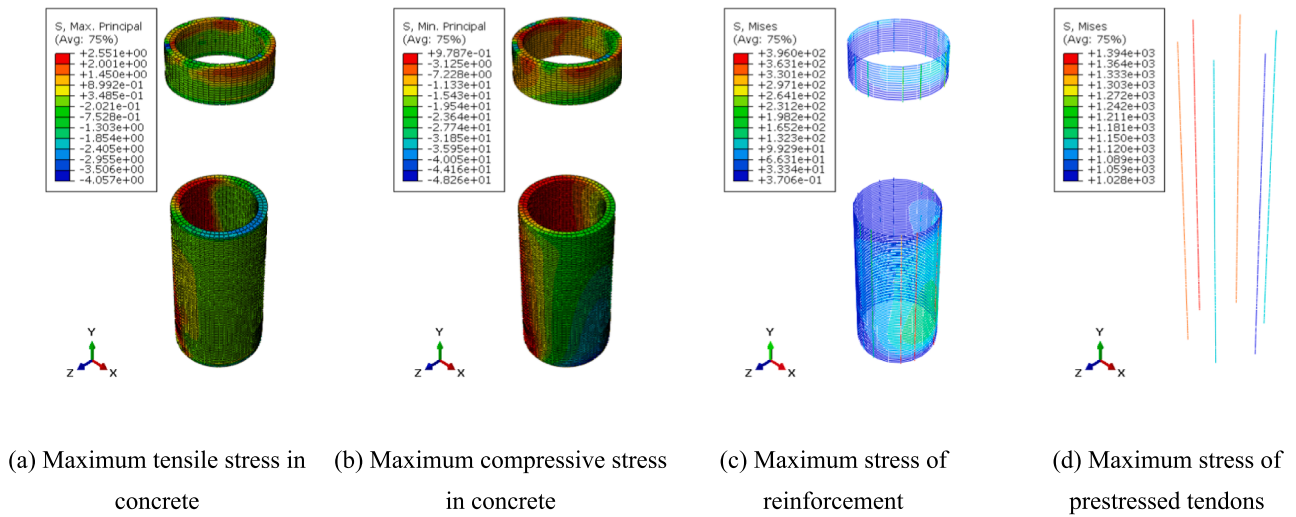


Fig. 13. Stress nephogram of the two-scale model.

Table 7
Stress comparison of the three models.

Model	Solid element model	Two-scale model	Fiber beam model
Maximum tensile stress in concrete (MPa)	2.53	2.55	6.39
Relative Error (%)	–	0.79	152.57
Maximum compressive stress in concrete (MPa)	–50.29	–48.26	–33.36
Relative Error (%)	–	4.04	33.66
Maximum Mises stress of non-prestressed reinforcement (MPa)	394.7	396.0	385.2
Relative Error (%)	–	0.33	2.41
Maximum Mises stress in prestressing tendon (MPa)	1,393	1,394	1,361
Relative Error (%)	–	0.07	2.30

bottom of the specimen. The bending cracks derived from the different models are consistent with the actual cracking patterns in the tested specimen which presents horizontal flexural cracks on the tensile side of the PC tube as shown in Fig. 3.

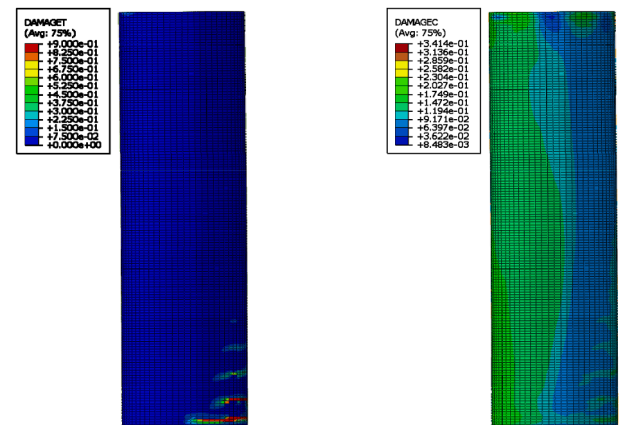


Fig. 14. Damage distribution of the solid element model.

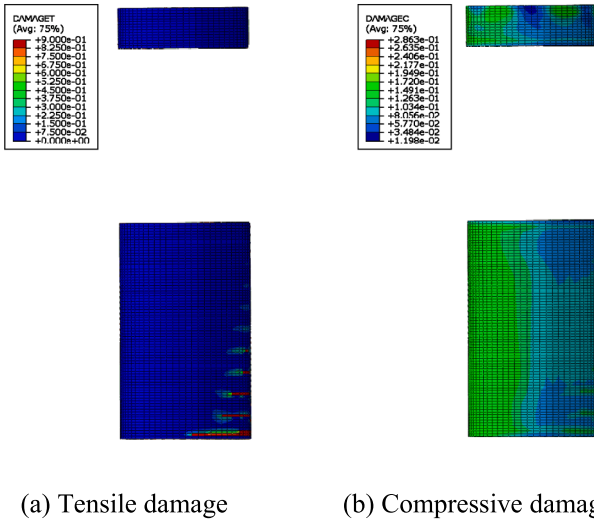


Fig. 15. Damage distribution of the two-scale model.

4.2.5. Computational efficiency comparison between different models

The numerical simulation for all the three numerical models is based on a DELL desktop computer with 16-core 32 processes and 32G memory. To exhibit the numerical simulation efficiency of the three different models, the comparison of the element numbers and the corresponding simulation time between the three numerical models are presented in Table 8.

It can be observed from Table 8 that the solid element model has the largest number of elements and takes the longest computation time, while the fiber beam model has the least number of fiber elements and computation time. The analysis time of the two-scale model was greatly reduced due to the reduction of the total element number when compared with that of the model with solid elements. The two-scale models are qualified of providing both global and local behavior of the tested PCSH specimen with acceptable accuracy and a shorter computation time.

5. Two-scale numerical simulation on a full-scale optimized PCSH wind turbine tower and the comparison with sole solid and fiber element models

After validating the effectiveness of the two-scale models described above, in this section, the two-scale modelling approach is utilized to study the global and local behavior of an optimized full-scale PCSH wind turbine tower [31,32]. The rated power of the wind turbine is 2 MW and the rotating speed in the rated power is 17 rpm. The rotor diameter is 93.4 m. The cut-in, rated, and cut-out wind speeds are 3 m/s, 11 m/s, 25 m/s, respectively. The weight of the wind turbine is 128.5 t and the distance from the gravity center of the wind turbine to the center of the tower is 3.7 m. The dimensions of the PCSH wind turbine tower are listed in Table 9. The PC segment of the PCSH tower is connected to the steel section through a steel flange with a thickness of 90 mm. The global and local behavior of the optimized full-scale PCSH model is simulated with the solid element model, two-scale model, and fiber beam model for comparison. The detailed material parameters are listed in Tables 10 and 11.

Table 8
Element number and calculation time.

Model	Numbers of elements	CPU time/s
Solid element model	36,096	140,328
Two-scale model	15,642	84,202
Fiber beam model	31	70

Table 9
PCSH wind turbine tower parameter.

Dimension	Value
Thickness of steel segment (mm)	11
Top diameter of steel segment (mm)	2,686
Base diameter of steel segment (mm)	3,296
Length of steel segment (mm)	22,200
Thickness at the top of PC segment (mm)	285
Thickness at the base of PC segment (mm)	285
Length of the PC segment (mm)	55,300
Diameter at the top of PC segment (mm)	3,565
Diameter at the base of PC segment (mm)	6,000
Longitudinal reinforcement	1820
Prestressed duct number	36
Prestressed reinforcement	$4 \Phi^S 1 \times 7 (d = 21.6 \text{ mm})$
Prestressed reinforcement area (mm^2)	41,040

5.1. Different models for an optimized full-scale PCSH wind turbine tower

For comparison, the three models for the PCSH wind turbine tower, including a fiber beam model, a two-scale model, and a solid element model as shown in Fig. 16, are established. Post-tensioned tendons are unbonded to the surrounding concrete of the optimized full-scale PCSH wind turbine tower. Prestress losses including anchorage, friction, stress relaxation, creep, and shrinkage have been incorporated. The optimized full-scale PCSH wind turbine tower is divided into several uniform section beam elements with a length of about two meters but different cross-section areas for simulation [20]. The bottom of the model is fixed and the wind turbine on the top of the tower is regarded as a rigid body with a certain mass [33]. For the two-scale model and fiber beam model, the steel segment is modeled as beam elements for simplicity [34].

5.2. Natural frequencies and modal shapes

In this section, the first two fundamental natural frequencies and modal shapes of the optimized full-scale PCSH wind turbine tower structure are simulated with the above three models and compared. The first and the second fore-aft mode shapes of the three models are presented in Fig. 17. Table 12 presents the comparison of the first two frequencies determined by the three models. According to Fig. 17 and Table 12, the natural frequencies and modal shapes of the three models are close to each other, which testify that the proposed two-scale modeling approach for the full-scale PCSH tower can be used to determine the natural frequency and mode shape effectively with acceptable accuracy.

5.3. Structural analysis on the behavior of the full-scale PCSH tower with different models

To compare the stress distribution of the above three models, the stress nephograms of the PC segment of the solid element model and two-scale model are shown in Fig. 18 and Fig. 19 when the PCSH tower is under ULS as an example. The stress value and distribution of the fiber beam model can be reckoned by the section forces and moments with material mechanics and compared with the results of the solid element model and two-scale model. The wind direction is the x-direction. No collapse of the steel segment is found, and no separation occurs between the steel flange and the PC segment. The maximum compressive stresses of concrete of the three models are located near the connection between the concrete and flange at the compression side. The compressive stress distribution of the two-scale element model and solid element are similar and have stress concentration phenomenon which is difficult to be expressed by the fiber beam model. The maximum tensile stresses of concrete of the solid element model and the two-scale model are located near the connection at the tensile side while the maximum tensile stresses of concrete of the fiber beam model are located near the bottom of the tower at the tensile side. The reason for the difference is the fiber

Table 10

Material property of concrete of the optimized full-scale PCSH wind turbine model.

Material	Elastic modulus (MPa)	Density (kg/m ³)	Poisson's ratio	Compressive strength (MPa)	Tensile strength (MPa)	Dilation angles (°)	Eccentricity	K	Viscosity coefficient
C50	34,500	2,500	0.2	32.4	2.64	31	0.1	0.667	1.0e-5

Table 11

Material property of steel, reinforcement and prestressed tendon of the optimized full-scale PCSH wind turbine model.

Material	Elastic modulus (MPa)	Density (kg/m ³)	Poisson's ratio	Yield strength (MPa)	Tensile strength (MPa)
Q345	206,000	7,850	0.28	345	—
HPB300	200,000	7,850	0.3	300	420
HRB400	200,000	7,850	0.3	400	540
Strand	195,000	7,850	0.3	1,320	1,860

beam model cannot simulate the stress concentration phenomenon at the connection between the concrete and the flange. The maximum stress of the reinforcement of the three models are located near the connection at the compression side.

The displacement, maximum stress, and computation efficiency comparison of the three models are exhibited in Table 13. According to Table 13, the top displacements and the maximum stress in the reinforcement and prestressed tendons of the three models are similar. The maximum tensile and compressive stresses of concrete of the two-scale model are close to that of the solid element model. The errors of the tensile and compressive stresses of the concrete reckoned by the fiber beam model are large. Besides, the use of fiber beam elements reduces the number of elements. As the element number decreases, so does the computation time. The two-scale model can reflect the stress of the wind turbine tower at the critical position and accelerate the computing speed compared with that of the solid element model. The fiber beam model can reduce the calculation time as well, but it is difficult to obtain the structural stress at the critical position with complicated stress distribution intuitively. In addition, the result of the fiber beam model has

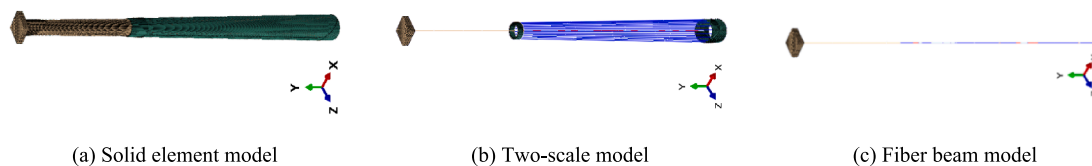
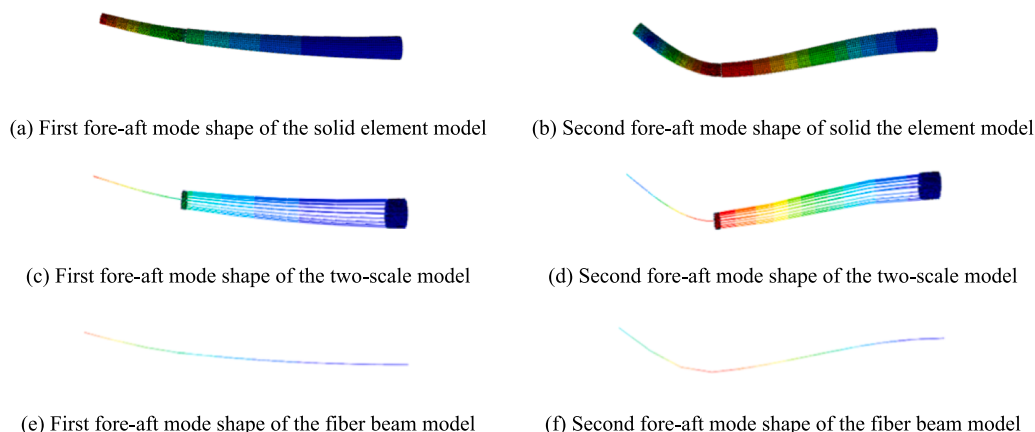
larger error due to the fact that the stress of the fiber beam model is estimated by the section internal force, which may be affected by the stress concentration phenomenon. Therefore, it is necessary to analyze and study the structural form and mechanical characteristics of critical locations using finer mesh. When only the global behavior of the wind turbine tower is considered, the PC segment is recommended to be simulated by the 3D fiber beam element. The solid element model can calculate the stress distribution of the PCSH tower, but it is time-consuming, and the stress in most positions is not essential.

To analyze the ultimate load-carrying capacity of the PCSH tower, the load acting on the top of the tower is simplified as a static transverse point load [35–37]. No separation between the flange and the PC segment is observed during the whole loading procedure. The load–displacement curves of different models of the PCSH tower are presented and compared in Fig. 20. According to Fig. 20, it can be concluded that the initial stiffness of the three models are similar and the PCSH towers show a ductile behavior. The load-carrying capacity and load–displacement curve of the two-scale model are close to that of the solid element model. As the loading proceeds, the stiffness degradation of the fiber beam model is worse than that of the solid element model

Table 12

Natural vibration frequency comparison of the models.

Model	Solid element model	Two-scale model	Fiber beam model
First-order frequency (Hz)	0.563	0.556	0.539
Relative Error (%)	—	1.24	4.26
Second-order frequency (Hz)	2.07	2.13	2.13
Relative Error (%)	—	2.90	2.90

**Fig. 16.** Different FEM models for the full-scale optimized PCSH wind turbine tower.**Fig. 17.** Comparison of the first two mode shapes of the three numerical models of the optimized full scale PCSH tower.

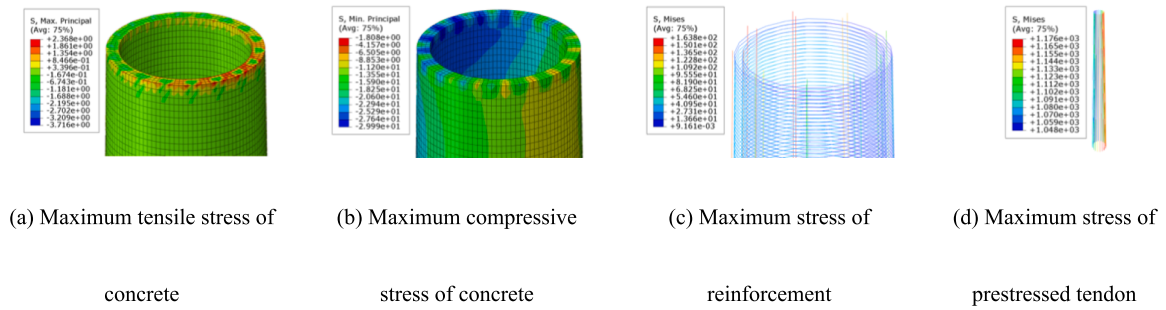


Fig. 18. Stress nephogram of the solid element model.

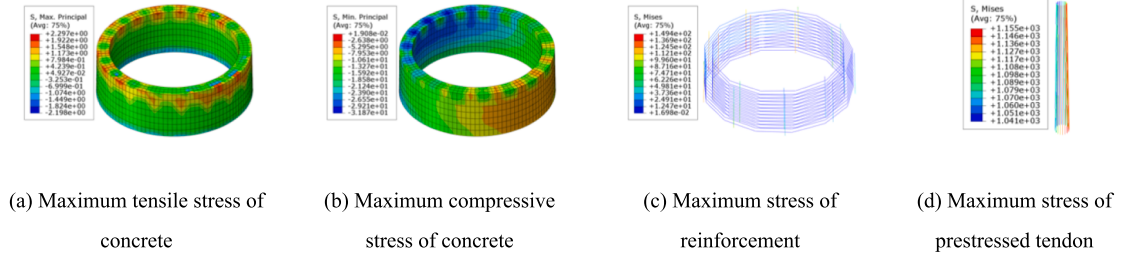


Fig. 19. Stress nephogram of the two-scale model.

Table 13

Displacement, stress and computation efficiency comparison of different models.

Model	Solid element model	Two-scale model	Fiber beam model
Top displacement (mm)	568.6	535.1	583.0
Relative Error (%)	–	5.89	2.53
Maximum tensile stress of concrete (MPa)	2.368	2.297	–1.326
Relative Error (%)	–	3.00	156.00
Maximum compressive stress of concrete (MPa)	–29.99	–31.87	–26.59
Relative Error (%)	–	6.27	11.34
Maximum stress of non-prestressed reinforcement (MPa)	163.8	149.4	174.7
Relative Error (%)	–	8.79	6.65
Maximum stress of prestressed reinforcement (MPa)	1,176	1,155	1094
Relative Error (%)	–	1.79	6.97
Element number	233,375	18,745	331
CPU time (s)	29,122	2,708	59

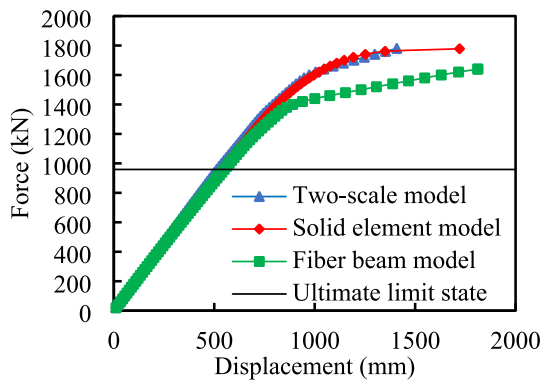


Fig. 20. Load-displacement curves of the three numerical models of the full-scale optimized PCSH tower model.

and the two-scale element model. The possible reason for the difference may be the fact that the stirrup cannot be considered in the beam elements, and it is difficult to precisely model the behavior of the critical parts of the optimized full-scale PCSH tower. Fig. 20 also depicts the load of the PCSH tower in the ULS, indicating that the structure can provide sufficient load capacity for the wind turbine tower. The capacity determined by the fiber beam model is lower than that of the solid element model and the two-scale model.

5.4. Fatigue of the full-scale wind turbine tower

Adopting the proposed two-scale approach, wind-induced fatigue analysis of the optimized PCSH wind turbine tower in section 5 is carried out. Dynamic response of the wind turbine towers corresponding to different wind speed under normal operational state is considered (between 3 and 25 m/s)[38]. Due to the long calculation time of solid element model, only the fiber beam model and the two-scale model are adopted here. Based on the rain flow method, the mean value and amplitude of third main stress and corresponding cycle number in one year at the bottom of the PC segment are illustrated in Fig. 21 as an example.

In the fatigue analysis, four critical regions, namely, the connection region between the steel tube and the flange, leeward side of the top and bottom region of the PC segment, and the prestressed tendons, are selected to carry out the wind-induced fatigue life analysis of the wind turbine tower. The fatigue reference compressive strength of concrete, the fatigue strength corresponding to the stress cycle number 2×10^6 and the minimum fatigue stress range of prestressing steel are 34 MPa, 69 MPa and 170 MPa, respectively. Adopting the time domain analysis method, the shortest fatigue life of different tower models is listed in Table 14[39]. According to Table 14, the fatigue life of each part of the towers is more than 20 years and meets the design requirements. The fatigue life on the top region of the PC segment is shorter than the bottom region, because the stress value of the top region is larger than that of the bottom region. The leeward side of the top region of concrete of the two-scale model has the shortest fatigue life. This is because the influence of prestressed reinforcement leads to the stress concentration phenomenon on the top of the concrete. Because it is difficult for the fiber beam model to simulate the stress concentration of the concrete,

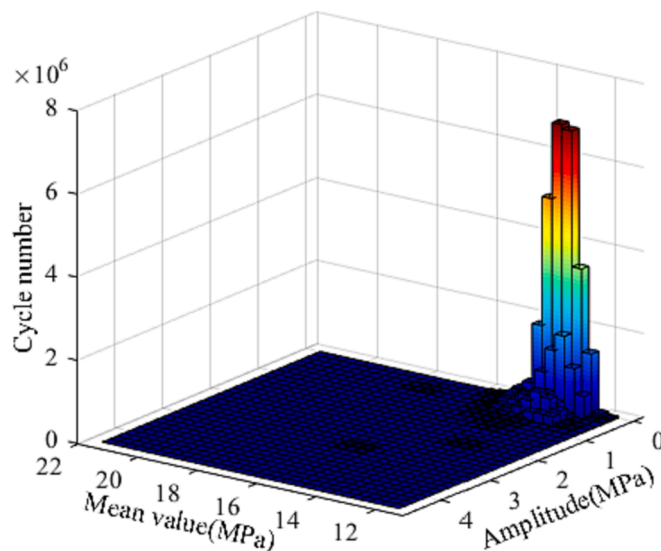


Fig. 21. The mean value and amplitude of stress and corresponding cycle number.

Table 14
The fatigue life calculation results.

Fatigue life	Connection region between the steel tube and the flange	Top region of the PC segment	Bottom region of the PC segment	Prestressed tendons
Two-scale model (year)	65.19	26.13	6.74×10^6	6.26×10^{10}
Fiber beam model (year)	107.78	1,368	7.23×10^5	5.56×10^{18}

the fatigue life calculated by the fiber beam model is longer than the result of two-scale model, which may arise security risks.

6. Conclusions

In this paper, to investigate both the global and local mechanical behavior of an optimized full scale PCSH wind turbine tower efficiently, a two-scale finite element modeling approach is proposed. Based on both the experimental and numerical simulation results, the following main conclusions can be made.

1) To establish the two-scale model for the optimized full-scale PCSH wind turbine tower, a 3D fiber beam element is developed at first for the PC segments of the PCSH tower. With the developed 3D fiber beam elements, a two-scale modeling approach for the substructure of a scaled PCSH tower is established and both global and local behavior of the specimen under the ULS and SLS are numerically investigated with the two-scale models. The experimental results of the substructure of the scaled PCSH model indicate that the designed specimen can meet the design requirements under the ULS and SLS with redundancy. The developed two-scale model is validated with the test results of the substructure of the scaled PCSH tower and the comparison of the numerical results of the two-scale model with those of the solid element model and the fiber beam model also show the accuracy and efficiency of the developed two-scale model.

2) The developed fiber beam model can simulate the global behavior of the PCSH wind turbine tower but is hard to simulate the stress concentration phenomenon and the local failure mechanism at critical locations of the substructure of the scaled PCSH tower. Moreover, with the increase of the external force, the fiber beam model may become

inaccurate due to the complex stress in the connection section and the bottom of the tower and the occurrence of cracks, which are hard to be modeled in fiber elements. The fiber beam model can be used for predicting only global, but not local, behavior. Compared with the solid element model, the two-scale model can not only significantly reduce simulation time but describe the local failure behavior in critical parts of the PCSH tower. The numerical simulation results show that the steel–concrete connection section and the bottom of the PCSH tower are the critical locations where stress concentration and concrete cracking may occur.

3) Aiming at the static behavior and modal parameter of the full-scale PCSH wind turbine tower optimized by the authors in the previous study, the validated two-scale modeling approach is employed. The first two order frequencies and the corresponding modal shapes of the optimized PCSH tower are analyzed by the use of the two-scale model and the results are compared with those of the solid element model and the fiber beam model. Results show the developed two-scale model can be used to determine both the natural frequencies and model shapes and the static load–displacement curves with accuracy and high efficiency.

4) The proposed method can be used to calculate the fatigue of PCSH wind turbine tower. The leeward side of the top region of the PC segment has the shortest fatigue life of the PCSH tower because of the stress concentration. The top region of the PC segment is the critical location of the PCSH wind turbine tower in the fatigue analysis. Because it is difficult for the fiber beam model to simulate the stress concentration, the fatigue life calculated by the fiber beam model may arise security risks.

CRediT authorship contribution statement

Zeyu Li: Data curation, Writing – original draft, Visualization, Software, Investigation. **Bin Xu:** Conceptualization, Methodology, Supervision, Writing – review & editing, Funding acquisition. **Jiang Wang:** Software, Validation, Data curation. **Hongbing Chen:** Software, Validation, Visualization. **Hanbin Ge:** Methodology, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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